

GANESH PRASATH K R

Software Engineer | Backend Microservices | Agentic RAG Systems | Workflow Automation

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PROFESSIONAL SUMMARY

Software Engineer with 1+ year of production experience designing and shipping end-to-end AI-driven systems — from multi-tenant RAG chatbot platforms and biometric SaaS to compliance microservices. Specializes in Python, FastAPI, LangGraph, pgvector, and Docker, with a track record of building systems that go beyond the brief: adding SaaS multi-tenancy, security hardening, and production observability before they are asked for. Recognized with a Company Spot Award (Dec 2025) for technical vision and end-to-end ownership.

TECHNICAL SKILLS

AI / LLM	LangGraph, n8n, Ollama models, Gemini API, OpenAI-compatible, RAG pipelines
Vector & DB	PostgreSQL + pgvector, Qdrant, Meilisearch, Redis, MySQL, Firebase
Backend	Python 3.11/3.12, FastAPI, SQLAlchemy 2.0 async, Alembic, REST API design, microservices
ML / Vision	AdaFace IR-101 ONNX, MiniFASNet, Azure OCR, nomic-embed-text
DevOps	Docker, Docker Compose, Prometheus, Grafana, Traefik v3
Security	JWT, bcrypt, Fernet encryption, HMAC-SHA256, double-hashed API keys, replay protection, RLS

WORK EXPERIENCE

Software Engineer — AI & Backend | Ai4Solutions

May 2025 – Present

- Designed and delivered 5 production-grade systems end-to-end: a multi-tenant RAG chatbot SaaS, a self-hosted cloud storage platform, two AML compliance engines, and a biometric verification SaaS — all individually owned from architecture to deployment.
- Architected agentic backend workflows using LangGraph and n8n to orchestrate FastAPI microservices with LLM layers, including a three-tier retrieval pipeline (FAQ → Semantic Cache → RAG) that prevents unnecessary LLM calls.
- Engineered a dual-pool Gemini model rotation system with thread-safe quota tracking, cutting API cost to \$0/month and saving ~\$1,200/year while serving 3,000+ requests/day.
- Introduced SaaS multi-tenancy (control-plane schema: tenants, bots, api_keys, deployment_targets) mid-build before the first customer shipped — an architectural pivot that made both platforms commercially viable.

KEY PROJECTS

Custom Bot — Multi-Tenant AI Chatbot SaaS Platform — *Python · FastAPI · LangGraph · pgvector · Redis · React · Docker · n8n · MinIO*

- Built a production-grade, multi-tenant "Build Your Own Chatbot" SaaS (11 Docker services) enabling any company to create bots, upload knowledge, and deploy embeddable widgets — comparable to Intercom AI + Botpress, fully self-hosted.
- Implemented a three-tier retrieval pipeline — FAQ golden-source (with synonym normalization and token overlap scoring) → privacy-aware semantic cache → full RAG — reducing LLM calls and preventing cached personalized responses from leaking to other users.
- Designed double-hashed API keys (SHA-256 prefix lookup + bcrypt brute-force resistance), Shadow DOM widget embedding with origin validation, and HMAC-SHA256 webhook signatures with replay-attack protection — same patterns used by Stripe and GitHub.
- Shipped full observability: 50+ automated tests (pytest + Playwright E2E), load-testing scripts for 250 concurrent users, and per-service architecture docs with changelog and roadmap.

Axenza Drive — Multi-Tenant Cloud Storage SaaS with AI-Native Sync — *Python · FastAPI · PostgreSQL · Redis · MinIO · Meilisearch · Next.js 15 · Traefik v3*

- Built a commercial-grade, self-hosted OneDrive alternative (8 FastAPI microservices) with a defining differentiator: automatic real-time AI knowledge sync — files uploaded to a connected folder update the RAG chatbot's knowledge base with zero manual re-upload.
- Designed a 3-tier sharing model (public / private / connector-scoped) with Fernet-encrypted connector credentials at rest, HMAC-SHA256 webhook signatures (300-second replay window), and connector-bound bearer tokens that make shared URLs useless without matching credentials.
- Implemented recursive CTE folder scope resolution for infinitely nested subfolder sync, incremental vs full sync modes, and sequence-number-based event ordering to prevent out-of-order deletes from corrupting the knowledge base.
- Enforced dual-layer tenant isolation: PostgreSQL Row-Level Security as a second defense plus per-tenant MinIO credentials — a single application bug cannot cause cross-tenant data leakage.

AML Background Verification System — Compliance Microservices — *Python · FastAPI · n8n · Gemini API · Redis · Docker · Azure MySQL*

- Architected a 6-service KYC/AML screening platform across global sanctions lists, adverse media, and financial crime databases with an async non-blocking gateway and 6-tier fuzzy name-matching engine.
- Reduced API cost to \$0/month via dual-pool Gemini model rotation with thread-safe quota tracking; auto-generated PDF compliance reports with Ollama/Mistral AI advisory layer.

FaceAuth — Multi-Tenant Biometric Verification SaaS — *Python · FastAPI · AdaFace ONNX · MiniFASNet · MediaPipe · Qdrant · PostgreSQL · Redis*

- Launched multi-tenant SaaS biometric platform from scratch serving as both internal product and third-party REST API with per-tenant configurable fraud thresholds
- Designed hybrid liveness detection pipeline — passive multi-scale fusion and active challenge-response with MediaPipe 478point landmarks — paired with AdaFace IR-101 ONNX for 512-D face embeddings

AML Adverse Media Screening — Multilingual Compliance Pipeline — *Python · FastAPI · Ollama · MADLAD-400 · Redis · PostgreSQL · Docker · n8n*

- Constructed 6-microservice multilingual AML pipeline for Singapore fintech (MAS TRM compliant), routing searches natively across 4 search engines with LLM-generated name variants covering 9 Unicode scripts
- Achieved fully on-premise LLM risk scoring (zero PII leaving server) across 10 AML risk categories with relevance-weighted confidence aggregation .

EDUCATION

B.E. in Electronics and Communication Engineering — 80%

2024

Vel Tech High Tech Dr. Rangarajan Dr. Sakunthala Engineering College

AWARDS & RECOGNITION

Spot Award — Ai4Solutions (December 2025)

Awarded by CTO and Managing Director for exceptional technical vision, hands-on leadership, and end-to-end ownership of the Crescent Exchange mobile project.